

Introduction to Relational Query Languages

RELATIONAL CALCULUS MOTIVATION



Set of Tuples

Set-former notation:

Example

$$\left\{ p.name, p.price \middle| Product(p) \wedge \exists m \left(\begin{array}{c} Manufacturer(m) \wedge \\ p.manufacturer = m.name \wedge \\ m.state = 'TX' \end{array} \right) \right\}$$

Product

name	price	manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper

Manufacturer

name	address	city	state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

RELATIONAL ALGEBRA MOTIVATION

Familiar: Algebra on Numbers

$$(x + 3) - (x \times y)$$

- Variables & values
- Operations
- Equivalence properties

Algebra on Relations

Operations include

- Filtering for tuples meeting condition
- Joining two tables
- Set operations

Equality properties such as commutativity



In Practice

SQL core = RC + RA

SQL query optimization based upon RA and equality properties.